

Course 5

Privacy, Confidentiality

Privacy – data / information

= relation between collecting and distributing data

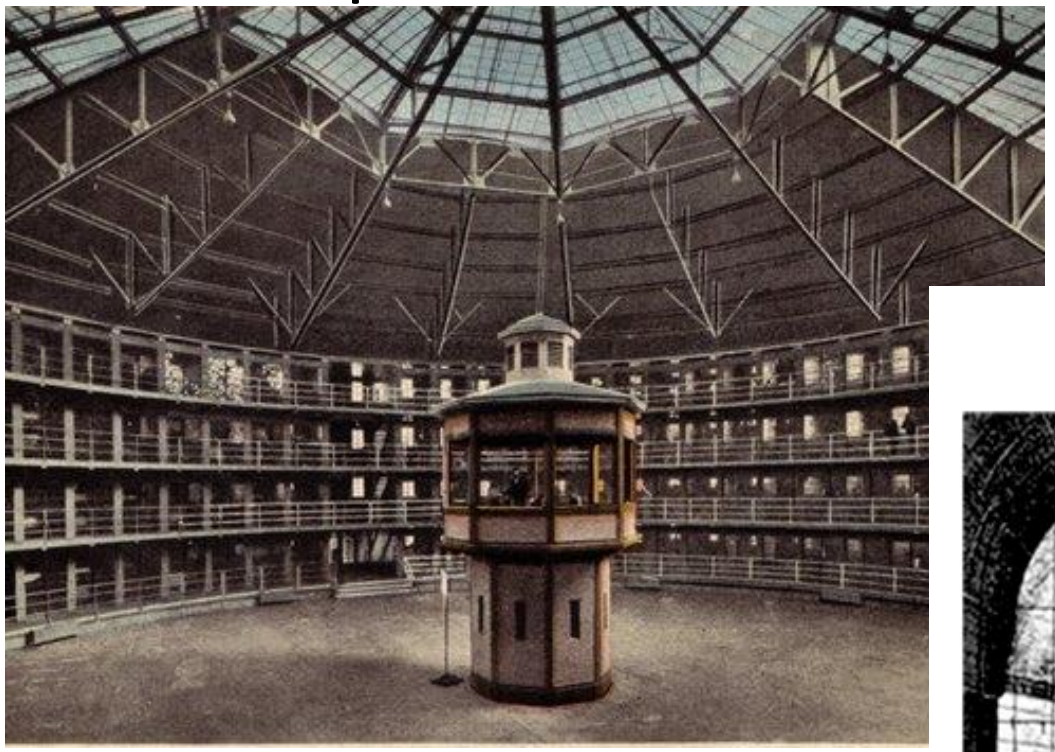
= freedom from unauthorized inclusion

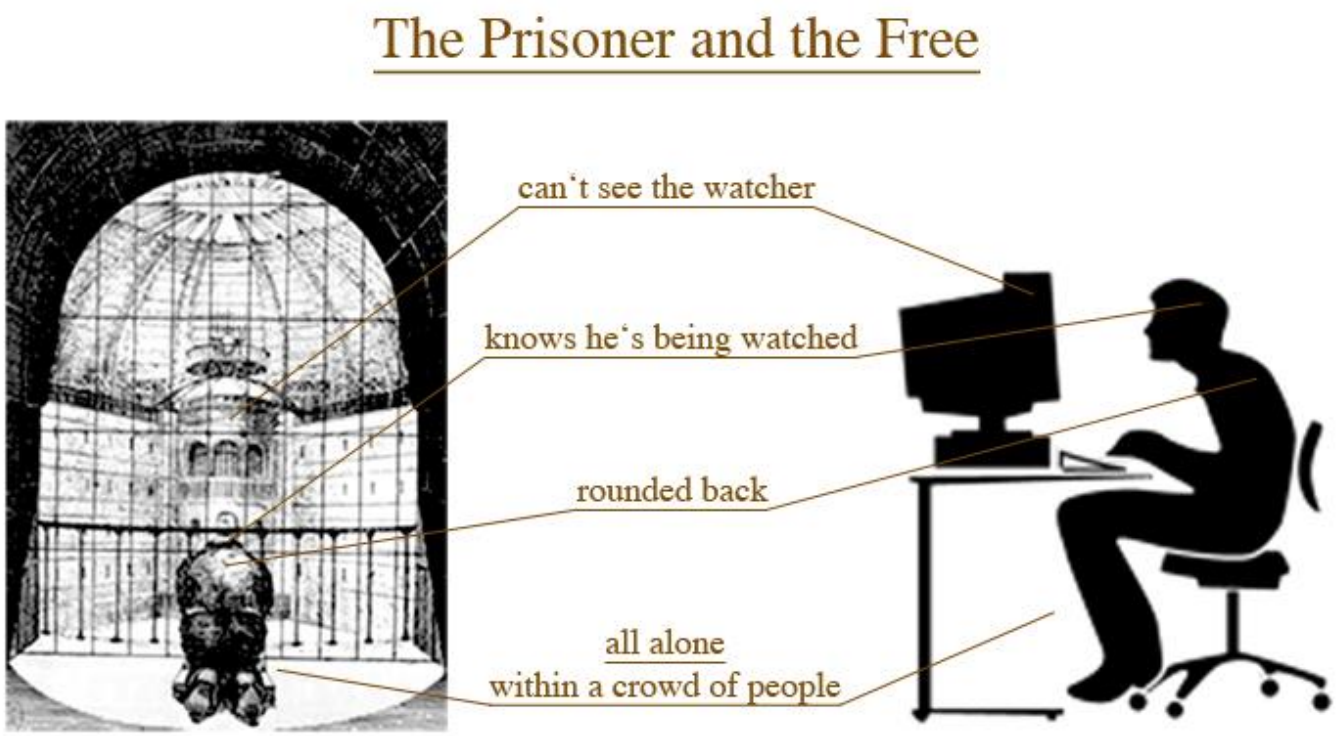
- Information - personal identification
- Sensitive Information
- WHAT?:
  - Medical
  - Financial
  - Identity (including residence)
  - Educational

Confidentiality

= **state** of keeping or being kept private

Panopticon




The Prisoner and the Free
can‘t see the watcher
knows he‘s being watched
rounded back
all alone
within a crowd of people

Facets of privacy

- Quality of life
- Society at large
- Professional conduct

- Transparent vs. Opaque
- Evidence based: Yes or No

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FEATURES

Feb 27, 2018, 10:25 PM GMT+2

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Palantir has secretly been using New Orleans to test its predictive policing technology

Palantir deployed a predictive policing system in New Orleans that even city council members don't know about

By **Ali Winston**
Illustrations by **Garret Beard** and **Alex Castro**

THE LOWDOWN

A Blog by Jonathan Low

Jun 16, 2019

Why Hong Kong Protesters Are Afraid To Use Their Metro Cards



And other cautionary tales from the surveillance state: if it's electronic, they will find you. Jon

Shibani Mahtani reports in the Washington Post:

Protests against a bill allowing extraditions to China were marked by a coordinated effort by demonstrators to leave no trace for authorities and their tracking systems. Protesters used only secure digital messaging apps and went completely analogue in their movements: buying single-ride subway tickets instead

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Clearview AI

Solutions Impact Resources Media Company CREATE A

CAPITOL RIOTS

JANUARY 6, 2021 | WASHINGTON D.C.

Millions watched as rioters stormed the United States Capitol. Around the country, several law enforcement agencies turned to Clearview AI to investigate and help identify those rioters. One agency in particular was the Miami Police Department who were able to forward 13 potential matches to the FBI.

*This is the **greatest threat we've faced in my lifetime.** The peaceful transition of power is foundational to our republic.*

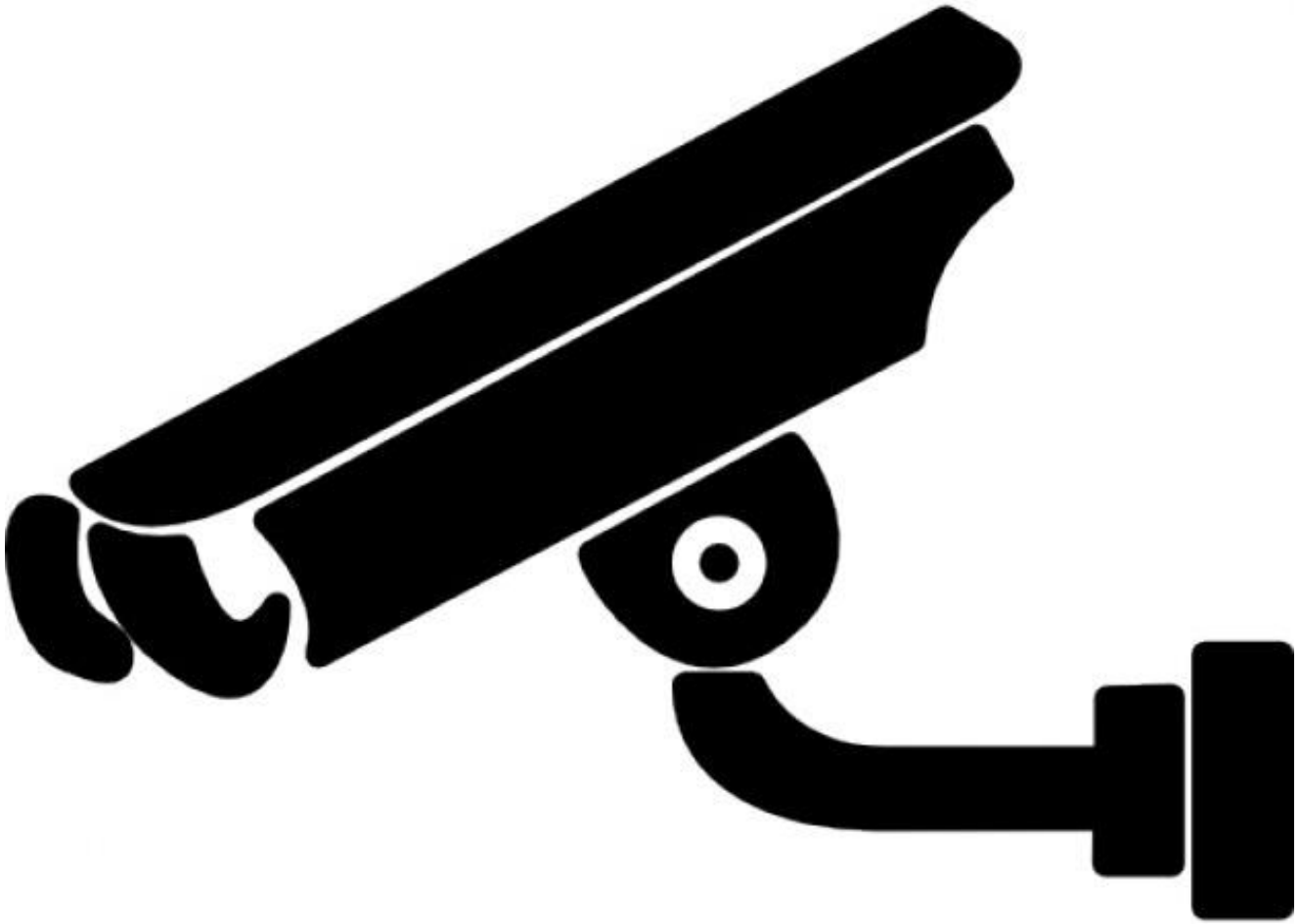


– ARMANDO AGUILAR
Assistant Chief, Miami PD

The New York Times

The facial-recognition app Clearview sees a spike in use after Capitol attack

Surveillance



Considerations

- Harms most vulnerable
- Data collected for one reason used for other purposes
- Error in data

https://urbanomnibus.net/2020/01/caught-in-the-spotlight/
https://www.fastcompany.com/90317495/another-tax-on-the-poor-surrendering-privacy-for-survival

Privacy – professional conduct

- **ACM Code**(1973):

“An ACM member, whenever dealing with data concerning individuals, shall always consider the principle of the individual's privacy and seek to:

  - *Minimize the data collected*
  - *Limit authorized access to the data*
  - *Provide proper security for the data*
  - *Determine the required retention period of the data*
  - *Ensure proper disposal of the data”*

- (1992) update
  "It is the responsibility of professionals to maintain the privacy and integrity of data describing individuals. This includes taking precautions to ensure the accuracy of data, as well as protecting it from unauthorized access or accidental disclosure to inappropriate individuals."

Guidelines for Ethics and Privacy in studies

- https://www.uu.nl/sites/default/files/QuickScan-Survey.docx

Confidentiality

- Data confidentiality - protection of data from **unauthorized** access and disclosure, including means for protecting personal privacy and proprietary information

- Confidentiality – set of rules that limits access or place restrictions

- Confidentiality – as part of Cybsersecurity

Senzitive information - confidential

- Legal
- Medical
- Banking

Confidentiality Agreement/ NDA

- Work e-mail

Case study

Case Study - Who Can Change Proprietary Source Code (source http://www.onlineethics.org)

- Derek worked at a small software company X and was the main contributor to a software system for customer services (main app of X); never signed an agreement for proprietary software
- Now, Derek is working in a large company of customer services Y. He realised that making minor modification to the system that he developed, he can improve his work significantly.

Q1: Legal?

Q2: Ethical?

Q3: You are the owner of company X. How you avoid this situation to happen?

Case 1b - cont

- The new company asked Derek to signed an agreement by which the software developed during his employement is the property of the company

Scenario 1: Other collegues will want to use the software? Should he give it to them?

Scenario 2: The management notice that he is using it, and asks him to do it for the whole company

Scenario 3: Imagine that in the meanwhile, he went out for a beer with friends, employees at company X, and told them about the software system. What should they do?

Case 2

- Maya is a data scientist at a health tech startup that collects anonymized user data from wearable fitness devices. While analyzing patterns to improve the app, she realizes that combining certain data points (like GPS paths and timestamps) will improve performance of the predictions but also she observed that it could indirectly reveal users' home addresses and daily routines.

- Her manager encourages her to use all available data to make predictions more accurate, but Maya feels uneasy about using data that could identify users even if it's not explicitly labeled as personal.

Q: What should Maya do?

Case 3: *"Cloud computing enhances privacy and confidentiality more effectively than traditional on-premise systems." [ChatGPT]*

Pro

- Cloud service includes encryption, multi-factor authentication, and intrusion detection systems
- Cloud services ensure controlled access and reliable backups, reducing the risks of data loss and unauthorized access
- Standard compliant

Con

- Loss of control: When data is stored in the cloud, users rely on third-party providers
- Global distribution of cloud servers – different countries
- Breach and Misconfiguration Risks

Q1: Who should be responsible for protecting data privacy in the cloud — the provider or the client?

Q2: Does cloud computing's scalability justify potential privacy trade-offs?

Case 4: *"Enhancing cybersecurity measures inherently improves privacy and confidentiality for individuals and organizations"[ChatGPT]*

Pro

- Cybersecurity frameworks use encryption and access management systems
- Cybersecurity tools (e.g., AI-based monitoring) detect and block intrusions before private data is exposed
- Trust and Compliance

Con

- Some cybersecurity measures (e.g., monitoring employee activity or network traffic) can infringe on personal privacy
- Security systems often collect large amounts of behavioral and usage data

Q1: Can cybersecurity measures be considered privacy-friendly if they involve constant surveillance or data tracking?

Q2: Should individuals sacrifice some privacy for stronger cybersecurity protections?